



Ph.D. Positions in Trustworthy and Multi-Agent GeoAI

Group of Dr. Peng Luo · Incoming Assistant Professor, August 2026

School of Earth, Environment, and Sustainability · University of Iowa

I will join the School of Earth, Environment, and Sustainability (SEES) at the University of Iowa as a tenure-track Assistant Professor in August 2026, and I am building a new research group at the intersection of Geographic Information Science (GIScience), Geospatial Artificial Intelligence (GeoAI), and spatial data science. I am recruiting two fully funded Ph.D. students to begin in Fall 2027. Each position is fully funded through a research or teaching assistantship, covering stipend, tuition, and benefits.

Research Directions

Position 1 - Trustworthy GeoAI. Developing reliable, interpretable, and uncertainty-aware geospatial AI. Topics include spatial explicit modeling, geospatial explainable AI (XAI), and uncertainty quantification of GeoAI models for spatial prediction and association analysis.

Position 2 - Multi-Agent GeoAI Modeling. Building LLM-driven, multi-agent frameworks for automated geospatial model discovery, reasoning, and scalable, self-evolving spatial analysis.

Qualifications

- A Bachelor's or Master's degree in GIScience, computer science, statistics, remote sensing, geography, or a closely related field.
- Strong programming skills (e.g., Python) and a genuine interest in spatial data science and AI.
- Self-motivated, curious, and able to communicate clearly in written and spoken English. Prior research experience is a plus but is not required.

What I Offer

- Full funding through a research or teaching assistantship (stipend, tuition, and benefits) for the duration of the Ph.D.
- Close, individualized mentorship and a clear path toward an academic or research career.
- An active international collaboration network and support for conference travel, computing resources, and publications.

Mentoring Philosophy

My mentoring philosophy is grounded in a fundamental Kantian principle: always treat people as ends in themselves, never merely as means. Guided by this conviction, my hopes and expectations for our partnership follow a clear hierarchy.

First - self-discovery over output.

My foremost responsibility is not to maximize your external achievements or output, but to safeguard the space and opportunity for you to fully experience the doctoral journey. I have walked that road myself; I tasted both its sweetness and its hardship. Yet in the end, looking back, I realized that my greatest reward was a deeper understanding of who I am - and for that, I remain profoundly grateful for the journey. A

Ph.D. is, first and foremost, a process of coming to know yourself on your own terms.

Second - intellectual freedom and the good taste it cultivates.

Built upon that foundation, I hold your intellectual freedom as paramount. I hope you will discover the geographic questions that truly fascinate you and, in the best case, develop enough healthy intellectual audacity to believe that you are the one uniquely positioned to care about - and ultimately solve - a particular problem. I believe that in the age of AI, a researcher's taste will matter more than ever. It simply makes no sense to publish many average papers; a single piece of work with genuine taste outweighs them all. And I am convinced that such taste can only emerge when you truly care about the questions themselves - only when you are doing what genuinely fascinates you will you produce work of real taste.

Third - strategic alignment and shared growth.

To be entirely transparent: as a junior faculty member, I navigate the real pressures of academic survival and promotion. While the first two principles always take precedence, I hope your interests can intersect with my own research trajectory. When our paths align, I can offer my most expert guidance with full confidence, and we both thrive.

How to Reach Me

If you are interested, please email me directly at peng-luo@uiowa.edu with your CV, transcripts, and a brief statement of research interests (indicate which position interests you), using the subject line "Prospective Ph.D. - [your name]." I review inquiries on a rolling basis and will follow up with promising candidates for a conversation. I encourage reaching out before submitting a formal application.

Formal application: Admitted students enroll through the Ph.D. in Geography program (GIScience track) at SEES. The University's application deadline is December 1 for the following Fall. I will guide selected candidates through the process - please contact me first.

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